

1. Project Overview

Project title: Online Retail EDA: Sales Patterns, Customer Behavior, and Cancellations

Business context

An online retailer wants to understand its transaction patterns before deciding which products, customer groups, and markets deserve closer attention. Exploratory data analysis can reveal concentrations in sales, purchasing patterns, cancellations, and unusual transactions that may affect business reporting.

Project objective

Explore historical retail transactions using Python to assess data quality, identify meaningful patterns, and develop evidence-based recommendations for further investigation.

Intended stakeholder

A hypothetical e-commerce operations or commercial manager.

Decisions this analysis could support

Identify products and markets to review, customer groups to investigate for retention opportunities, and cancellation patterns that warrant operational follow-up.

Analysis approach

Use Pandas to inspect and prepare the data, then use Matplotlib and Seaborn to explore distributions, trends, and relationships. Document processing decisions and assess whether unusual transactions materially change the findings.

Successful project outcome

A reproducible notebook containing documented quality checks, clearly labeled charts, supported interpretations, and a concise business summary. Recommendations will distinguish observed findings from hypotheses requiring additional data or testing.

Scope

This is an independent portfolio project using historical transaction data. Calculated sales values will not be presented as profit. Recommendations will not be described as implemented business improvements.

2. Exploratory Questions

1. What does each row represent, and how many records, distinct invoices, products, and identified customers are present?
2. Which fields contain missing values, repeated records, or unusual values? How should these issues be handled for different analyses?
3. How are quantities, unit prices, line-item values, and order values distributed? Are a small number of unusually large transactions driving the results?
4. How do sales values and order counts vary over time? Are the reporting periods complete enough for fair comparisons?
5. Which products and countries contribute the largest shares of sales, and how concentrated are those contributions?
6. Among customers with recorded identifiers, how frequently do customers purchase, and how much sales value comes from repeat purchasers?
7. What patterns appear in cancellation records, and how do including or excluding them affect sales summaries?
8. Which findings remain consistent when unusual transactions are examined separately, and what business actions or further investigations do those findings support?

3. Data Source

Dataset: Online Retail

Provider: UCI Machine Learning Repository

Creator: Daqing Chen

Source URL: <https://archive.ics.uci.edu/dataset/352/online+retail>

Original filename: Online Retail.xlsx

Source-reported size: 541,909 records and eight listed columns. These will be independently verified in Python.

Reporting period: December 1, 2010, through December 9, 2011, according to the source.

Business context: Historical transactions from a UK-based online retailer selling gifts. The source notes that many customers are wholesalers.

License: Creative Commons Attribution 4.0 International (CC BY 4.0). The project will credit the dataset creator and source and describe any processing changes.

Dataset citation: Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5BW33>

Initial limitations: This dataset represents one historical retailer and should not be treated as representative of all e-commerce businesses. It does not provide product costs needed to calculate profit. The reporting window includes partial months, which must be considered when comparing monthly results. Missing values and other quality issues will be checked independently rather than relying solely on source metadata.

4. Data Preparation and Quality

The original Excel dataset contained 541,909 invoice-line records and eight columns. It was loaded using Pandas in Google Colab, with invoice, product, and customer identifiers preserved as text. The original data remained unchanged.

Quality checks identified 1,454 missing descriptions, 135,080 missing customer IDs, and 5,268 exact repetitions beyond their first occurrence. All records were retained in the prepared dataset. Missing customer IDs were excluded only from customer-level analysis.

Records were classified as merchandise candidates, special/non-merchandise entries, or unresolved. Postage, fees, accounting adjustments, discounts, samples, and vouchers were separated from merchandise analysis. Thirteen unresolved records were retained for review but excluded from merchandise subsets.

Positive merchandise entries required a merchandise-candidate code, positive quantity and price, and no cancellation prefix. Eligible merchandise cancellations required a merchandise-candidate code, a C-prefixed invoice, negative quantity, and positive price.

Line value was calculated as $\text{Quantity} \times \text{UnitPrice}$. Positive merchandise value totaled £10,271,404.55. Eligible cancellation magnitude totaled £478,724.18, producing signed merchandise value of £9,792,680.37. These measures do not establish profit or recognized revenue.

Exact repetitions were flagged rather than automatically removed. Sensitivity tests assessed their influence and the influence of two large transaction/cancellation pairs. A separate investigation examined a picnic-basket code with differing descriptions and an unusually large line value.

Monthly, product, and country summaries reconciled with the baseline. Customer summaries reconciled with their identified purchasing subset. December 2011 was marked as incomplete because the dataset ends on December 9.

5. Exploratory Analysis

Seven analyses were completed using Pandas and Matplotlib:

1. Merchandise baseline and distributions: examined quantities, prices, and line values, identifying a right-skewed distribution.
2. Sensitivity analysis: compared baseline totals with scenarios excluding exact repetitions, two large reversal pairs, or both.
3. Monthly patterns: compared positive and signed merchandise values and invoice counts, explicitly marking incomplete December coverage.
4. Product contribution: ranked products by signed value and tested ranking sensitivity. Investigated the picnic-basket code separately after identifying inconsistent descriptions and dependence on one large line.
5. Country contribution: measured UK concentration and compared leading non-UK countries.
6. Customer purchasing: assessed customer-ID coverage, observed repeat purchasing, and concentration among leading identified customers.
7. Cancellation patterns: compared overall and monthly cancellation-to-positive-value ratios with and without the two large reversal pairs.

The notebook contains the code, outputs, and exploratory charts. The Analysis Tracker records findings, quality decisions, field definitions, and chart details.

6. Key Findings

1. A small number of unusual records materially affects reporting.

Two large cancellation entries account for 51.31% of eligible merchandise cancellation magnitude. Excluding their matching positive and cancellation entries changes the cancellation-to-positive-value ratio from 4.66% to 2.32%, while leaving signed merchandise value unchanged.

Separately, excluding one £38,970 picnic-basket line changes that product code's signed-value rank from 9th to 168th. These findings demonstrate why aggregate totals and rankings require record-level investigation and sensitivity testing.

2. Identified repeat purchasers contribute most customer-cohort value.

Customers with at least two positive merchandise invoices represent 65.27% of the identified purchasing cohort and contribute 93.50% of its signed merchandise value. The top 10 customers contribute 16.54%.

Customer IDs cover 85.30% of positive merchandise value. These findings apply to observed identified purchasing customers, rather than the entire customer base, and do not measure retention or campaign effectiveness.

3. Recorded merchandise value is concentrated geographically.

The UK contributes 84.77% of overall signed merchandise value. Netherlands and EIRE are the leading non-UK countries, contributing £283,479.54 and £259,663.46, respectively.

Their small identified purchasing-customer counts—nine and three—warrant customer-concentration investigation. Country totals include eligible records without customer IDs, so these counts alone do not establish how broadly value is distributed.

7. Business Recommendations

1. Strengthen merchandise reporting and exception review.

Report positive merchandise value, cancellation magnitude, and signed merchandise value separately. Provide a clearly labeled sensitivity comparison for unusually large reversals.

Introduce an exception-review process for large line values, inconsistent product descriptions, unusual prices, and repeated records. Prioritize verification of picnic-basket code 22502 before using its ranking for inventory decisions.

Suggested monitoring measures include the number and value of flagged records, unresolved exceptions, and differences between baseline and sensitivity results.

2. Investigate repeat and high-value customer accounts.

Observed repeat purchasers contribute 93.50% of identified customer-cohort signed value. Review their purchasing recency, service needs, margins, and cancellation activity before selecting accounts for retention initiatives.

Improve customer-ID capture so customer reporting covers more eligible transactions. If a retention initiative is tested, compare outcomes with an appropriate comparison group rather than attributing every subsequent purchase to the campaign.

Suggested measures include customer-ID coverage, subsequent purchasing within a defined follow-up period, and contribution margin where cost data becomes available.

3. Review geographic concentration before allocating expansion resources.

Monitor the UK's 84.77% contribution to signed merchandise value. Investigate how international value is distributed among identified customers, particularly in Netherlands and EIRE.

Combine transaction findings with fulfillment costs, margins, acquisition costs, and market information before recommending expansion. High recorded value alone does not establish a profitable growth opportunity.

Suggested measures include country contribution, customer concentration, and country-level margins when available.

4. Investigate the late-year increase before using it for planning.

Both positive invoice counts and signed merchandise value increased from September through November, with November producing the highest complete-month value.

Examine the contributing products, customers, promotions, and stock availability. Obtain additional years before treating the pattern as recurring seasonality. Exclude incomplete December 2011 from direct full-month comparisons.

These recommendations are proposed actions from an independent portfolio analysis. No implementation or measured business improvement is claimed.

8. Limitations

Historical scope and generalizability

The dataset represents one UK-based retailer from December 1, 2010, through December 9, 2011. Findings should not be generalized to current e-commerce businesses without additional evidence.

Incomplete reporting period

December 2011 contains only part of the month. One complete December–November cycle is insufficient to establish recurring annual seasonality.

Missing customer information

Customer IDs are missing from approximately 24.93% of original records. Identified customers account for 85.30% of eligible positive merchandise value, so customer findings cover only that observed subset.

Repeated records

The dataset contains 5,268 exact repetitions beyond their first occurrence. Without a unique line-item identifier, their validity cannot be confirmed. They remain in baseline analyses, with exclusions tested separately.

Product classification and identity

Merchandise eligibility uses documented code and transaction rules. Thirteen unresolved records were excluded from merchandise subsets. Different code spellings remain separate, and descriptions are display labels rather than verified product-master definitions.

The picnic-basket code contains differing descriptions and prices. Its high ranking depends heavily on one transaction whose business meaning remains unconfirmed.

Cancellation interpretation

Cancellation reasons and explicit original-order links are unavailable. Cancellation entries may relate to purchases outside a particular month or reporting window. The cancellation-to-positive-value ratio is not an order cancellation rate.

Customer behavior

Repeat purchasing is defined using distinct positive invoices observed within the window, including invoices that may later be canceled. Customers have unequal observation periods. This is not a lifetime customer measure or retention rate.

Financial and causal boundaries

The data does not provide product costs, fulfillment costs, acquisition costs, or margins. Recorded merchandise values do not establish profit or recognized revenue.

The analysis identifies descriptive patterns. It does not establish their causes, predict campaign effectiveness, or demonstrate implemented business outcomes.

9. Portfolio Links

GitHub repository:

<https://github.com/chetollyercoleman/online-retail-eda>

EDA notebook:

https://github.com/chetollyercoleman/online-retail-eda/blob/main/notebooks/Online_Retail_EDA.ipynb

Charts:

<https://github.com/chetollyercoleman/online-retail-eda/tree/main/figures>

Summary tables:

<https://github.com/chetollyercoleman/online-retail-eda/tree/main/summaries>